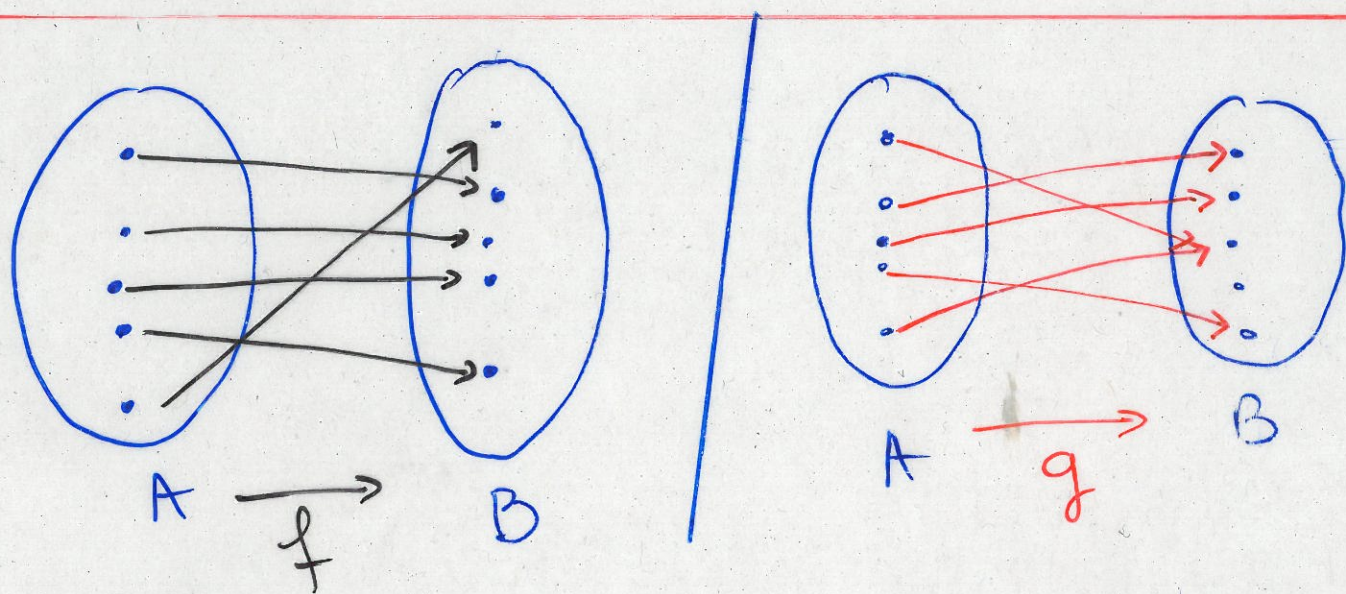


1.8 Inverse functions

Definition 1: A function f is called one-to-one function if it never takes on the same value twice; that is $f(x_1) \neq f(x_2)$ whenever $x_1 \neq x_2$

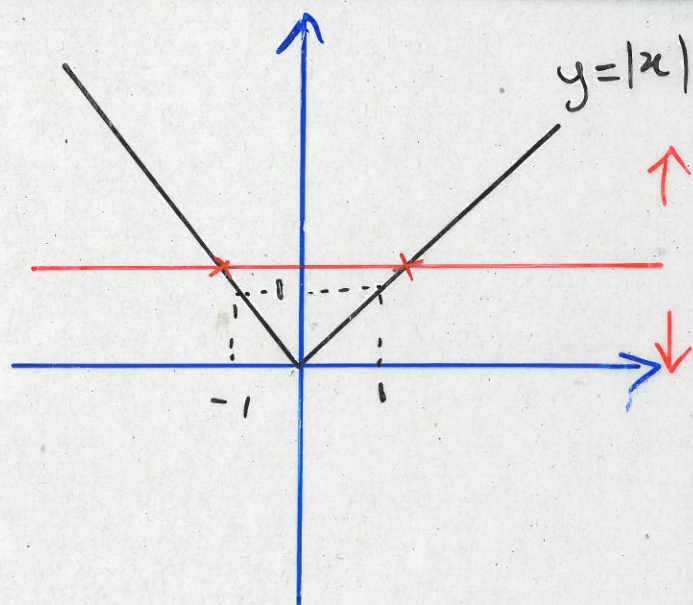
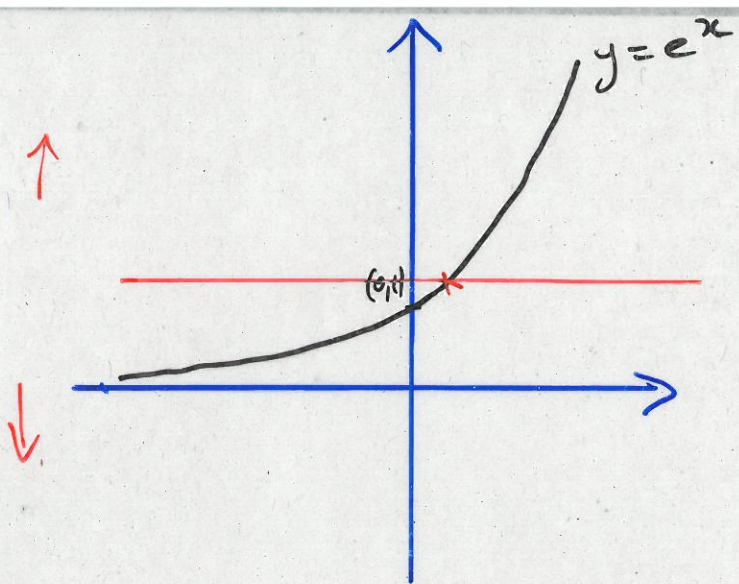


Horizontal line test: A function is one-to-one if and only if no horizontal line intersects its graph more than one.

Example 1: Determine if f is one-to-one

(a) $f(x) = e^x$

(b) $f(x) = |x|$



By horizontal line test (a) is one-to-one but (b) is not.

Definition 2: Let f be a one-to-one function with domain A and range B . Then its inverse function f^{-1} has domain B and range A and is defined by

$$f^{-1}(y) = x \iff f(x) = y \quad \text{for any } y \in B$$

$$\underline{\text{domain of } f^{-1} = \text{range of } f}$$

$$\underline{\text{range of } f^{-1} = \text{domain of } f}$$



$$f^{-1}(x) \neq \left(\frac{1}{f(x)} \right) \text{ or } [f(x)]^{-1}$$

Since " x " is, in general, used as the independent variable, Definition 2 is equivalent to

$$f^{-1}(x) = y \iff f(y) = x$$

Cancellation equations:

$$f^{-1}(f(x)) = x \quad \text{for every } x \text{ in } A$$

$$f(f^{-1}(x)) = x \quad \forall x \in B$$

How to find the inverse f^{-1} of a one-to-one f

(1) Write $y = f(x)$

(2) solve this equation for x in terms of y (if possible)

(3) to express f^{-1} as a function of x , interchange x and y . The resulting equation is

$$y = f^{-1}(x)$$

Example 2 : Determine $f^{-1}(x)$, given $f(x)$

(a) $f(x) = 1 + \sqrt{2+3x}$

(b) $f(x) = 2^x$

(c) $f(x) = x^4 + 1$, $x \geq 0$

Example 2 :

$$(a) \quad y = 1 + \sqrt{2+3x}$$

$$(y-1) = 2+3x$$

$$x = \frac{(y-1)^2 - 2}{3}$$

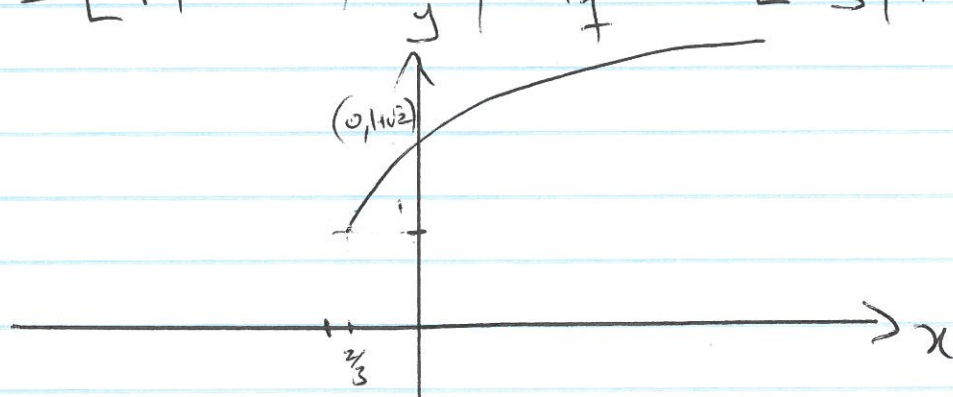
$$\Rightarrow f^{-1}(x) = \frac{x-3}{3} = \frac{x}{3} - 1$$

$$D_f = \{x \in \mathbb{R} / 2+3x \geq 0\} \\ = \left[-\frac{2}{3}, +\infty\right)$$

by H.C.T f is one-to-one

$$R_f = [1, +\infty)$$

$$D_{f^{-1}} = [1, +\infty), \quad R_{f^{-1}} = \left[-\frac{2}{3}, +\infty\right)$$



$$(b) \quad y = 2^x$$

$$D_f = \mathbb{R}, \quad R_f = (0, +\infty)$$

$$\log_2 y = x$$

$$\Rightarrow f^{-1}(x) = \log_2 x$$

$$(c) \quad y = x^4 + 1$$

$$D_f = [0, +\infty), \quad R_f = [1, +\infty)$$

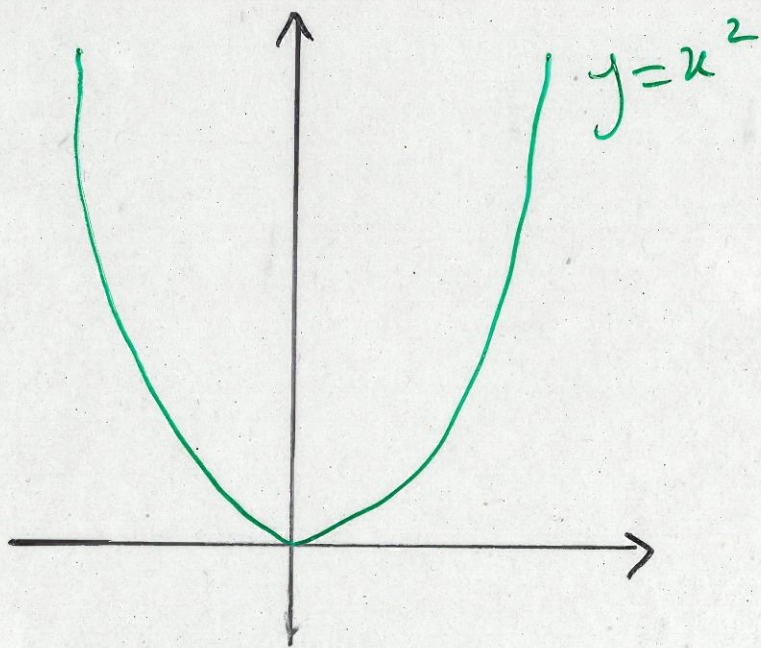
$$(y-1) = x^4$$

$$(y-1)^{\frac{1}{4}} = x \quad (\text{since } x \geq 0) \Rightarrow f^{-1}(x) = \sqrt[4]{x-1}$$

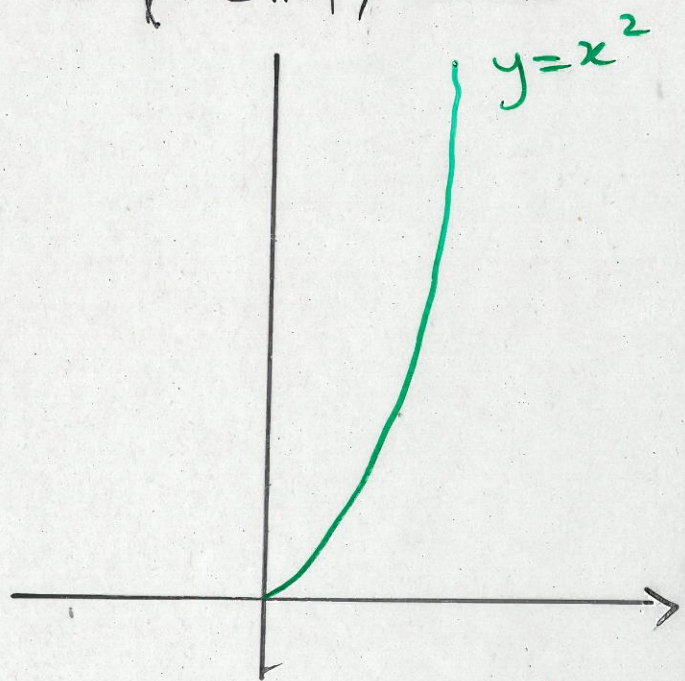
NB: $f(x) = x^2$ is not one-to-one
 but if $g(x) = x^2$, $x \geq 0$, that is
 $D_g = [0, \infty)$ then g is one-to-one
 g is the restriction of f on $[0, \infty)$
 and $g^{-1}(x) = \sqrt{x}$, $D_{g^{-1}} = R_g = [0, \infty)$
 $R_{g^{-1}} = D_g = [0, \infty)$

In this manner, we may find the
 inverse function of

$\sin x$, $\cos x$, $\tan x$, x^{2n} ($n \in \mathbb{N}$), $\dots$



$$D_f = (-\infty, \infty)$$



$$D_f = [0, \infty)$$